

ASSINIBOIA, SK, Canada

WMO: 714870

Lat: 49.730N

Lon: 105.950W

Elev: 726

StdP: 92.91

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-27.6	-34.2	0.2	-30.5	-30.9	0.2	-27.5	14.2	-7.3	13.0	-7.4	3.8	270	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	32.0	17.8	29.9	17.1	27.9	16.5	20.2	27.1	18.9	26.0	17.9	25.1	4.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	14.1	23.0	16.6	12.9	21.7	15.3	11.9	20.4	61.6	26.9	56.9	26.1	53.2	25.2	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	11.0	9.7	-34.1	35.8	3.1	1.9	-36.3	37.1	-38.2	38.3	-39.9	39.3	-42.2	40.7		
			-34.2	22.7	3.1	1.7	-36.4	23.8	-38.3	24.8	-40.0	25.8	-42.3	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-11.3	-10.1	-3.7	4.5	10.9	15.8	19.2	18.5	13.1	5.3	-2.9	-9.5
	DBStd	12.55	8.54	8.11	8.02	5.61	4.70	3.67	3.26	3.73	4.78	5.29	6.96	7.93
	HDD10.0	3050	662	564	424	176	46	3	0	0	22	161	389	603
	HDD18.3	5283	920	797	682	414	233	93	28	44	167	404	638	862
	CDD10.0	941	0	0	1	12	74	176	284	264	115	15	1	0
	CDD18.3	133	0	0	0	0	3	16	55	49	10	0	0	0
	CDH23.3	1967	0	0	0	7	84	225	696	727	218	9	0	0
	CDH26.7	690	0	0	0	1	17	59	249	289	74	1	0	0
Wind	WSAvg	4.7	5.1	4.8	5.0	5.0	5.1	4.6	3.9	3.9	4.3	4.8	4.9	4.9
Precipitation	PrecAvg	400	16	11	25	25	58	67	51	44	39	21	15	23
	PrecMax	535	43	29	57	105	163	171	170	173	115	67	27	55
	PrecMin	324	5	2	0	0	0	8	0	4	0	0	0	6
	PrecStd	60	10	7	15	26	40	39	40	40	30	18	9	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.8	16.9	24.2	29.0	31.4	34.9	35.1	32.2	24.7	16.7	7.5
		MCWB	2.8	4.2	8.6	11.3	14.5	17.5	18.8	17.7	16.6	12.6	9.0	3.2
	2%	DB	3.7	4.4	12.1	20.0	25.7	28.2	31.7	32.4	29.0	20.1	11.7	5.0
		MCWB	1.5	2.0	6.1	9.9	13.2	17.2	18.9	17.0	14.8	10.9	6.0	2.0
	5%	DB	2.4	2.5	8.9	17.2	23.0	25.8	29.5	30.0	25.7	16.5	8.3	3.1
		MCWB	0.7	0.8	4.8	8.7	12.2	15.7	18.5	16.6	14.1	9.2	4.2	0.9
	10%	DB	0.6	0.6	5.9	14.2	20.4	23.7	27.2	27.6	22.5	13.6	5.7	1.1
		MCWB	-0.7	-0.6	3.2	7.1	11.0	14.7	17.6	16.2	12.8	7.8	2.7	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	4.5	9.2	12.5	16.0	20.7	22.9	20.7	17.9	13.7	9.0	4.0
		MCDB	5.4	7.5	15.8	21.7	25.0	27.5	29.3	28.1	26.5	21.8	16.4	6.5
	2%	WB	1.7	2.2	6.7	10.6	14.2	18.5	21.0	19.2	16.1	11.5	6.4	2.3
		MCDB	3.4	4.0	11.1	18.5	22.4	25.1	27.7	26.9	25.5	18.2	10.9	4.4
	5%	WB	0.8	0.9	4.9	9.2	13.1	17.0	19.6	18.1	14.7	9.7	4.4	1.0
		MCDB	2.3	2.4	8.6	16.1	20.7	22.8	26.4	26.2	23.1	15.5	7.8	3.2
	10%	WB	-0.7	-0.5	3.1	7.6	12.0	15.8	18.5	17.1	13.5	8.3	2.8	-0.5
		MCDB	0.6	0.6	5.8	13.3	18.5	21.6	25.3	25.1	21.3	13.1	5.4	1.2
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.1	10.2	12.2	12.9	11.7	13.4	14.5	13.7	11.7	10.0	9.4
		MCDBR	10.3	10.8	13.5	17.4	17.3	15.5	16.8	18.7	19.2	16.8	13.6	10.5
		MCWBR	8.6	8.6	8.7	9.0	7.6	6.7	7.1	6.9	8.0	9.0	8.7	8.2
	5% WB	MCDBR	10.1	10.4	13.1	15.8	14.6	13.1	14.1	15.1	16.6	15.3	13.1	10.4
		MCWBR	8.5	8.5	8.7	8.6	7.0	6.5	7.0	6.5	7.6	8.6	8.6	8.3
Clear-Sky Solar Irradiance	taub	0.251	0.256	0.271	0.324	0.345	0.344	0.357	0.365	0.323	0.281	0.253	0.230	
	taud	2.402	2.439	2.529	2.399	2.385	2.438	2.414	2.365	2.451	2.555	2.521	2.472	
	Ebn at Noon	822	912	955	924	913	913	894	869	877	871	823	813	
	Edh at Noon	61	76	84	107	114	109	110	110	91	68	54	49	
All-Sky Solar Radiation	RadAvg	1.30	2.33	3.62	4.86	5.82	6.20	6.73	5.41	3.98	2.45	1.38	0.99	
	RadStd	0.13	0.17	0.24	0.33	0.43	0.38	0.39	0.38	0.36	0.28	0.17	0.09	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.77	N/A	N/A	N/A	N/A	
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page